

Problem Statement:

Given a sorted array and a value x , the floor of x is the largest element in array smaller than or equal to x . Write divide and conquer algorithm to find floor of x .

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int a[], int low, int high, int x) {
4      if (low > high)
5          return -1;
6
7      int mid = low + (high - low) / 2;
8
9      if (a[mid] == x)
10         return a[mid];
11
12     if (a[mid] > x)
13         return findFloor(a, low, mid - 1, x);
14
15     // a[mid] < x
16     int result = findFloor(a, mid + 1, high, x);
17
18     if (result == -1)
19         return a[mid];
20
21     return result;
22 }
23
24 int main() {
25     int n, x;
26
27     scanf("%d", &n);
28
29     int a[n];
30
31     for (int i = 0; i < n; i++) {
32         scanf("%d", &a[i]);
33     }
34
35     scanf("%d", &x);
36
37     int floor = findFloor(a, 0, n - 1, x);
38
39     printf("%d\n", floor);
40
41     return 0;
42 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓